

Senior Software Engineer — AI Systems & Infrastructures

Location: New York, NY (Hybrid)

Department: Intelligent Systems

Experience: Senior Level

ROLE OVERVIEW

Apex Systems Labs is seeking an exceptional Senior Software Engineer to architect, build, and optimize our next-generation document processing and intelligence platform. In this role, you will bridge the gap between heavy AI/ML capabilities and rock-solid backend infrastructure. You will design scalable data ingestion pipelines, integrate advanced NLP models, implement ultra-fast vector search architectures, and build intuitive full-stack user interfaces.

KEY RESPONSIBILITIES

- Architect and maintain high-throughput backend services using **FastAPI** and **Python**.
- Design and implement low-latency vector databases and semantic indexing systems using **pgvector** or specialized vector stores.
- Deploy, fine-tune, and optimize embeddings pipelines utilizing transformer models such as **Sentence-BERT** for document matching and retrieval-augmented generation (RAG).
- Build robust, responsive, and type-safe web dashboards and developer tools using **React** and **TypeScript**.
- Lead technical design, perform rigorous code reviews, and establish engineering best practices for scalable system architectures.

REQUIRED QUALIFICATIONS & TECHNICAL SKILLS

Core Programming: Python (Advanced), TypeScript, JavaScript, SQL.

Backend Frameworks: FastAPI, PostgreSQL (with heavy emphasis on pgvector or advanced query optimizations).

Machine Learning / NLP: Deep understanding of embeddings, transformer architectures (Sentence-BERT), text-processing, and semantic ranking systems.

Frontend Frameworks: React, modern state management, responsive web UI architecture.

Systems & Infrastructure: Scalable microservices, API contract design, performance tuning for data-heavy applications.

PREFERRED QUALIFICATIONS

- Experience working with OCR engines, PDF parsing libraries, and messy data extraction streams.
- Active contributions to open-source software packages or a strong portfolio of complex engineering side projects.
- Excellent written and verbal communication skills, with a track record of driving technical initiatives to completion.